

Chapter 1: Building the Database and Model

FullyStacked 

Login to your account

Email Address

you@somewhere.com

Password



Login

FullyStacked 

Start Workout

Today's Workout

DAY 1, WEEK 1

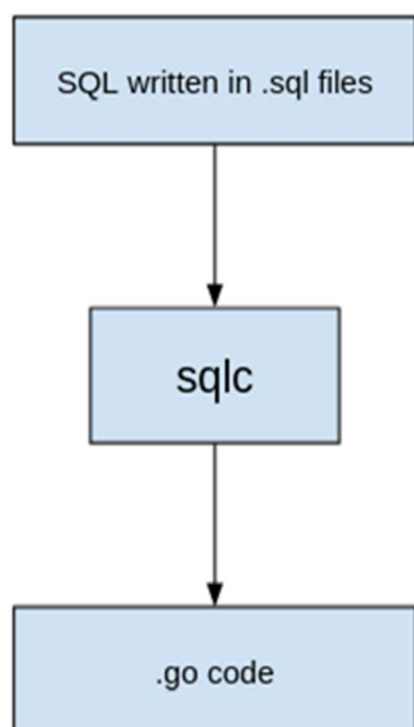
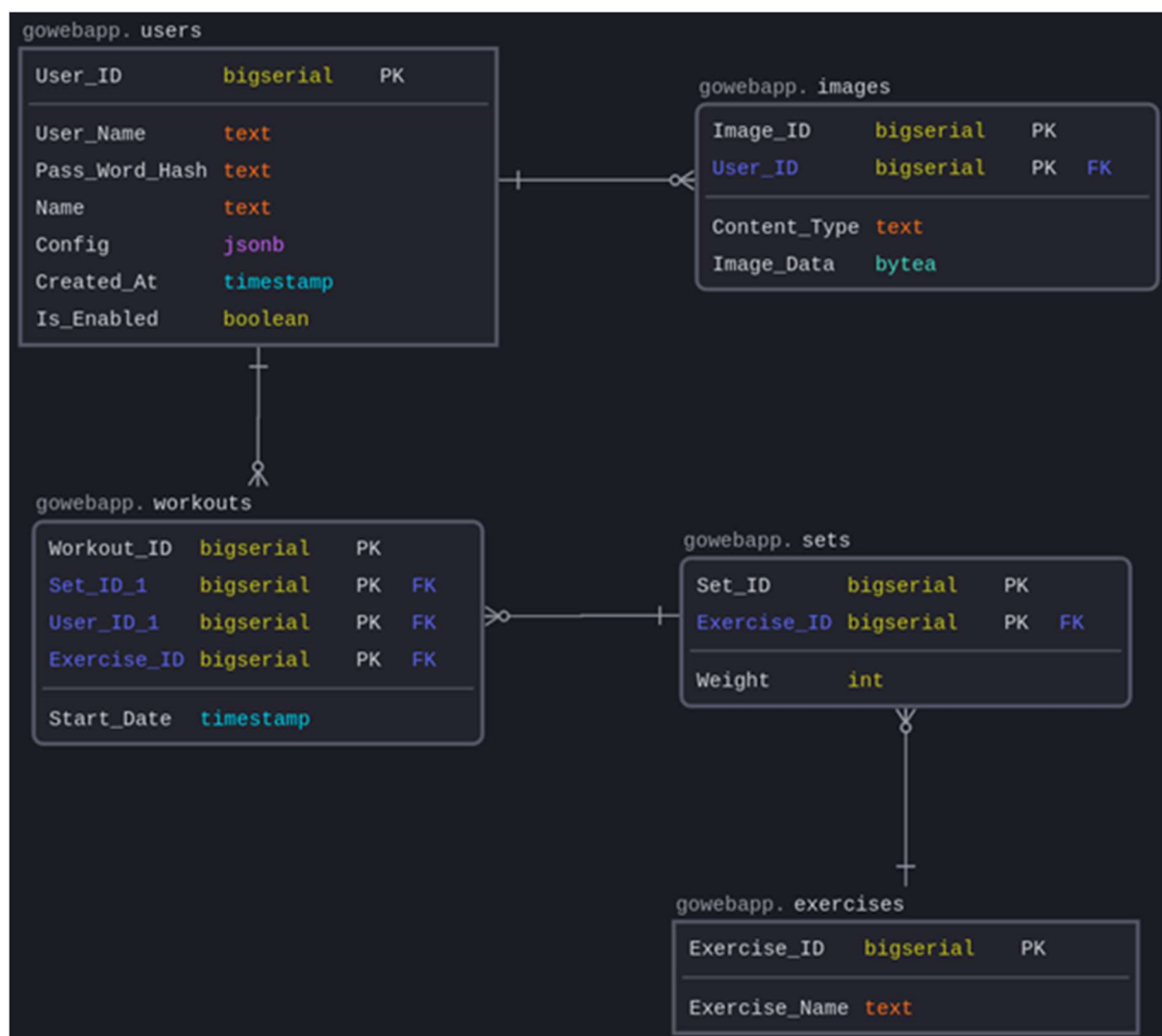
Bench Press	2x5 1x5+	100kg
Bench Press	2x5 1x5+	100kg
Bench Press	2x5 1x5+	100kg

Workout on 2/1/2021

DAY 1, WEEK 1

Bench Press	2x5 1x5+	100kg
Bench Press	2x5 1x5+	100kg
Bench Press	2x5 1x5+	100kg

No More Workouts



Chapter 2: Application Logging

2021/10/15 13:43 This is a raw message, no levels, no colors.

[INFO] 2021/10/15 13:43 This is an info message, with colors (if the output is terminal)

[WARN] 2021/10/15 13:43 This is a warning message

[ERROR] 2021/10/15 13:43 This is an error message

[DEBUG] 2021/10/15 13:43 This is a debug message

[FATAL] 2021/10/15 13:43 Fatal will exit no matter what

[DEBUG] 2021/10/17 19:15 Application logging to stdout = true

[INFO] 2021/10/17 19:16 Starting the application...

[INFO] 2021/10/17 19:16 Database connection fine

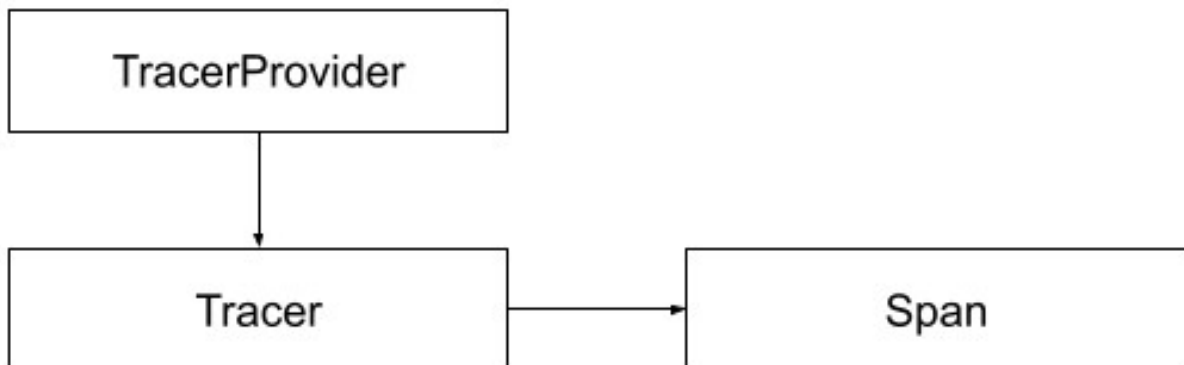
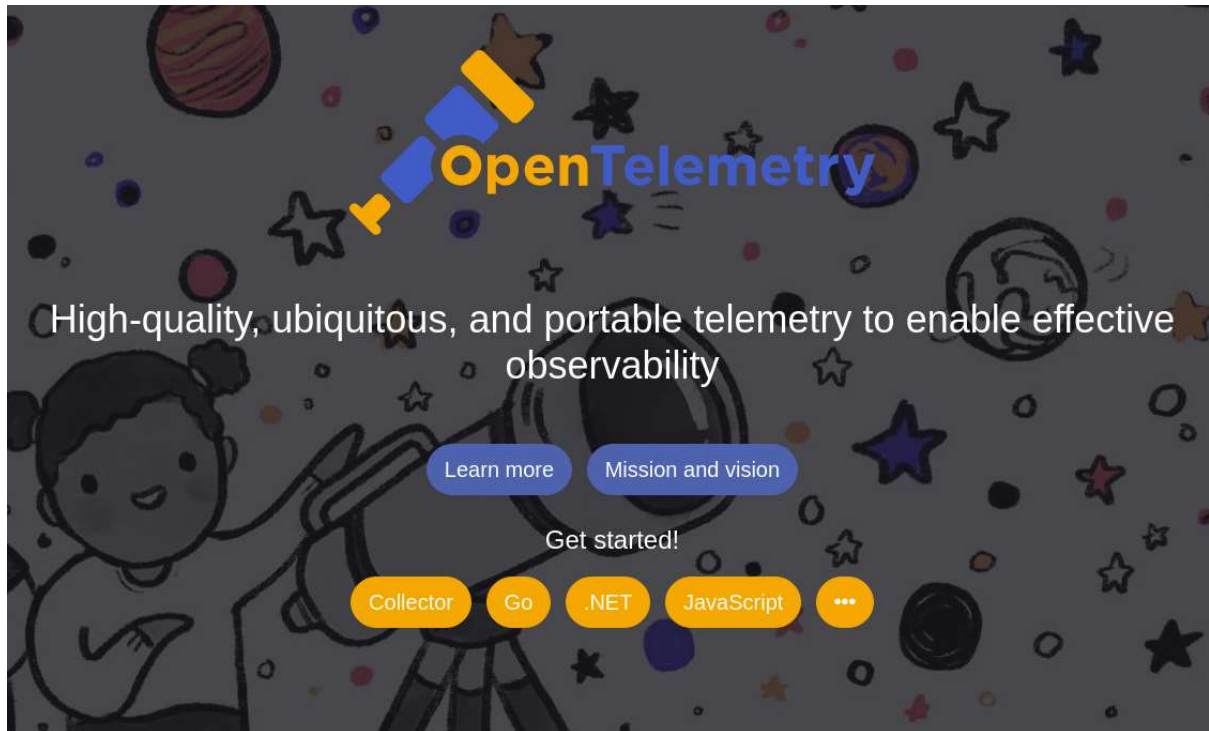
[INFO] 2021/10/17 19:16 Success - user creation

[INFO] 2021/10/17 19:16 Success - exercise creation

[INFO] 2021/10/17 19:16 Success - updating workout

[INFO] 2021/10/17 19:16 Application complete

Chapter 3: Application Metrics and Tracing



JAEGER UI
Search
Compare
System Architecture
About Jaeger

Search

JSON File

Service (0)

Select A Service

Operation (0)

all

Tags ?

http.status_code=200 error=true

Lookback

Last Hour

Max Duration

e.g. 1.2s, 100ms, 500us

Min Duration

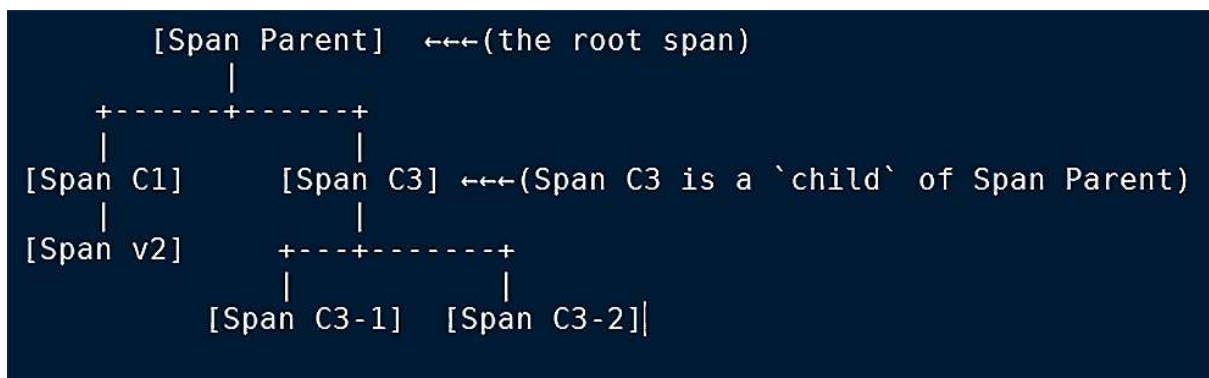
e.g. 1.2s, 100ms, 500us

Limit Results

20

Find Traces

Logo by Lav Polyakov
www.polyakovproduction.com



Search

JSON File

Service (2)

tracing

jaeger-query

tracing

all

Tags ?

http.status_code=200 error=true

Lookback

Last Hour

Max Duration

e.g. 1.2s, 100ms, 500us

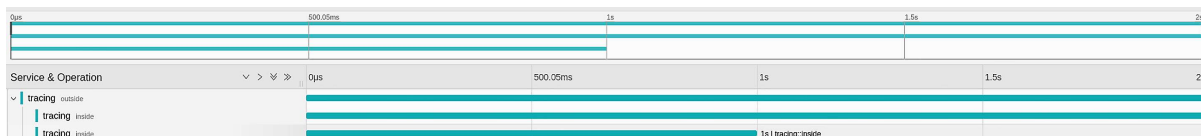
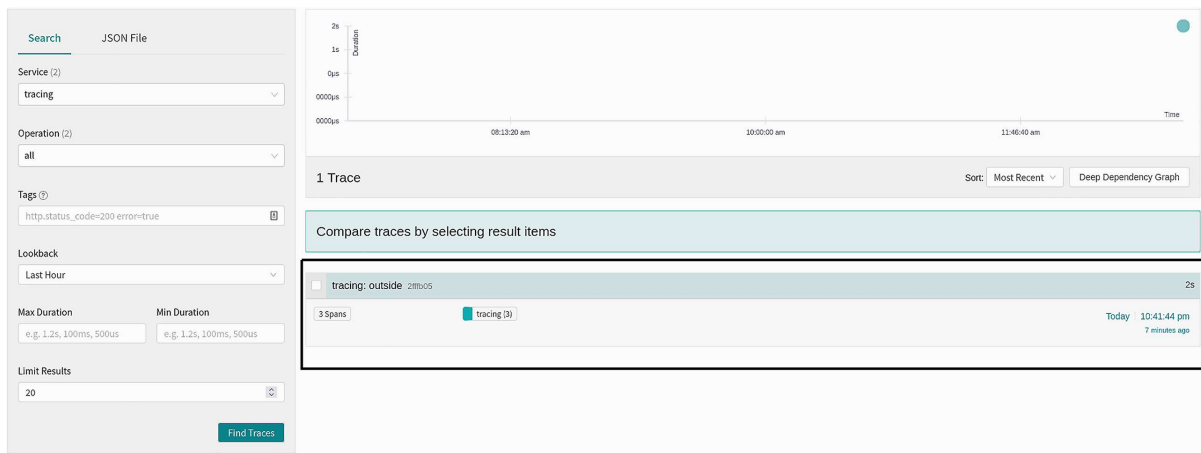
Min Duration

e.g. 1.2s, 100ms, 500us

Limit Results

20

Find Traces



The banner features the Prometheus logo and navigation links (DOCS, DOWNLOAD, COMMUNITY, SUPPORT & TRAINING, BLOG). The main text reads: "From metrics to insight" and "Power your metrics and alerting with the leading open-source monitoring solution." Below this are two buttons: "GET STARTED" and "DOWNLOAD".

🔍 Dimensional data

Prometheus implements a highly dimensional data model. Time series are identified by a metric name and a set of key-value pairs.

🔍 Powerful queries

PromQL allows slicing and dicing of collected time series data in order to generate ad-hoc graphs, tables, and alerts.

📈 Great visualization

Prometheus has multiple modes for visualizing data: a built-in expression browser, Grafana integration, and a console template language.

📦 Efficient storage

Prometheus stores time series in memory and on local disk in an efficient custom format. Scaling is achieved by functional sharding and federation.

⚙️ Simple operation

Each server is independent for reliability, relying only on local storage. Written in Go, all binaries are statically linked and easy to deploy.

⚠️ Precise alerting

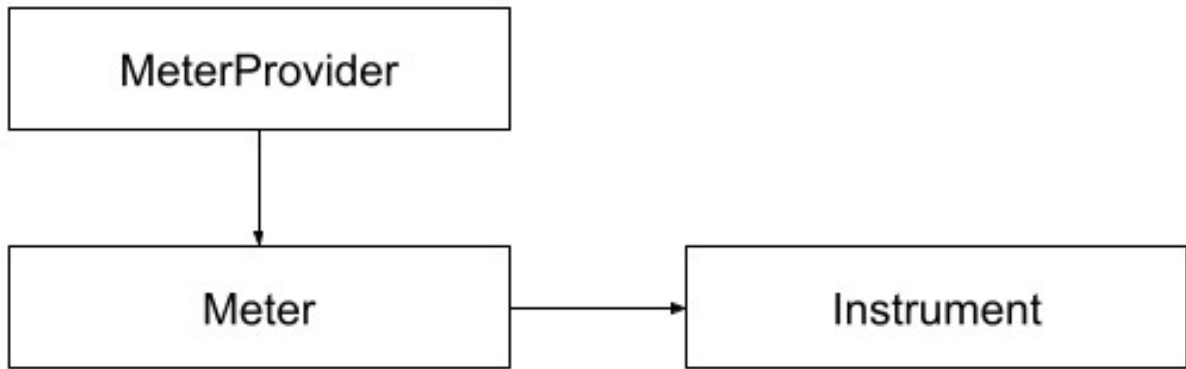
Alerts are defined based on Prometheus's flexible PromQL and maintain dimensional information. An alertmanager handles notifications and silencing.

🔗 Many client libraries

Client libraries allow easy instrumentation of services. Over ten languages are supported already and custom libraries are easy to implement.

🔗 Many integrations

Existing exporters allow bridging of third-party data into Prometheus. Examples: system statistics, as well as Docker, HAProxy, StatsD, and JMX metrics.



Prometheus Alerts Graph Status ▾ Help Classic UI

☒ Use local time ☐ Enable query history ☒ Enable autocomplete ☒ Enable highlighting ☐ Enable linter

🔍 Expression (press Shift+Enter for newlines) ⌕ Execute

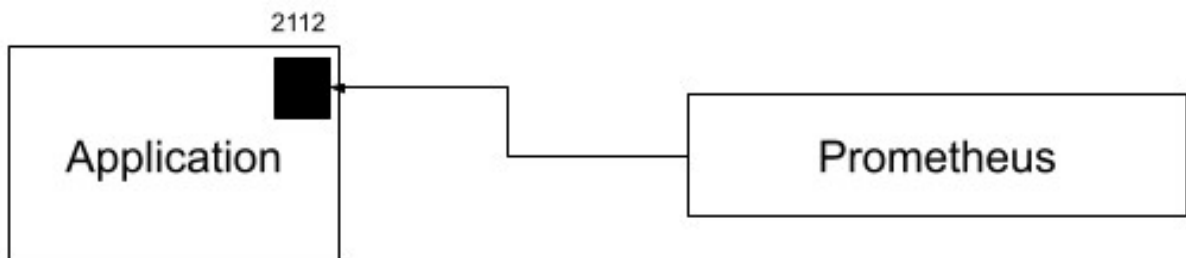
Table **Graph**

< Evaluation time >

No data queried yet

[Remove Panel](#)

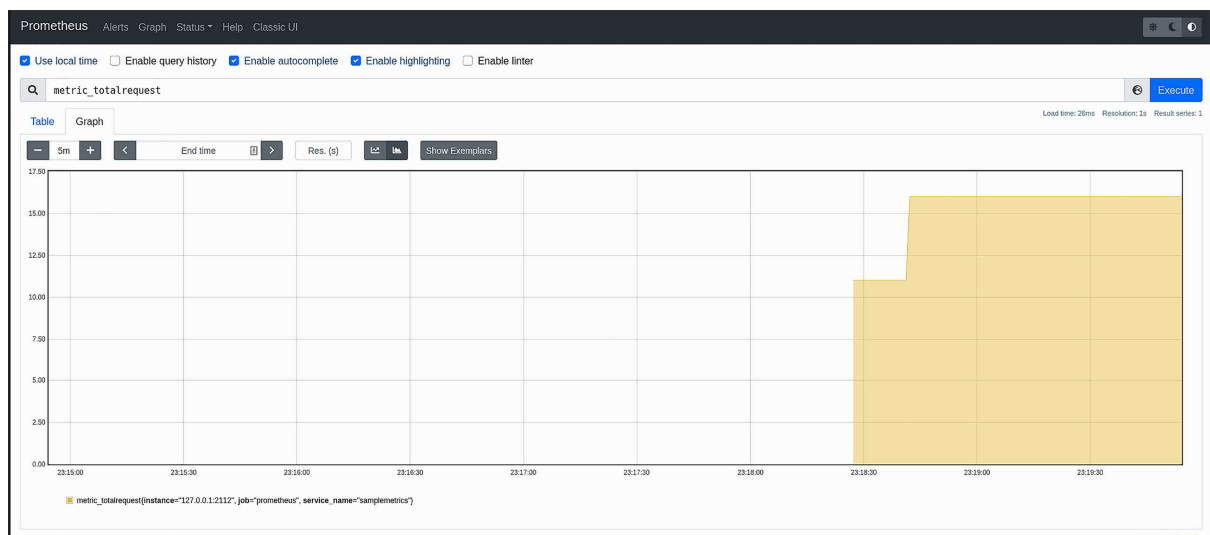
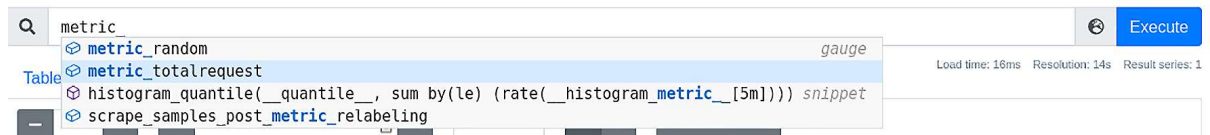
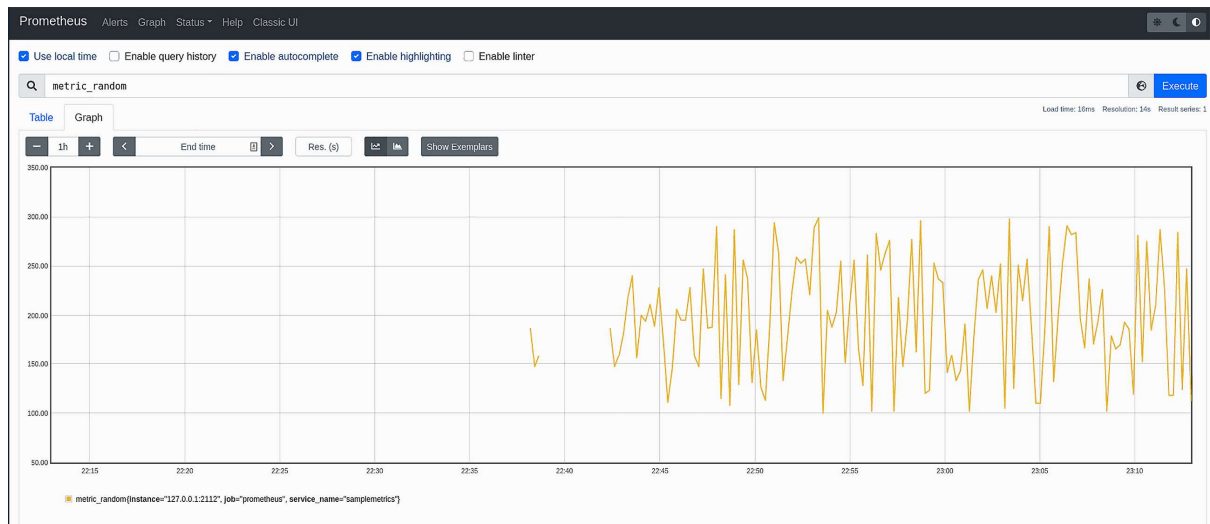
Add Panel



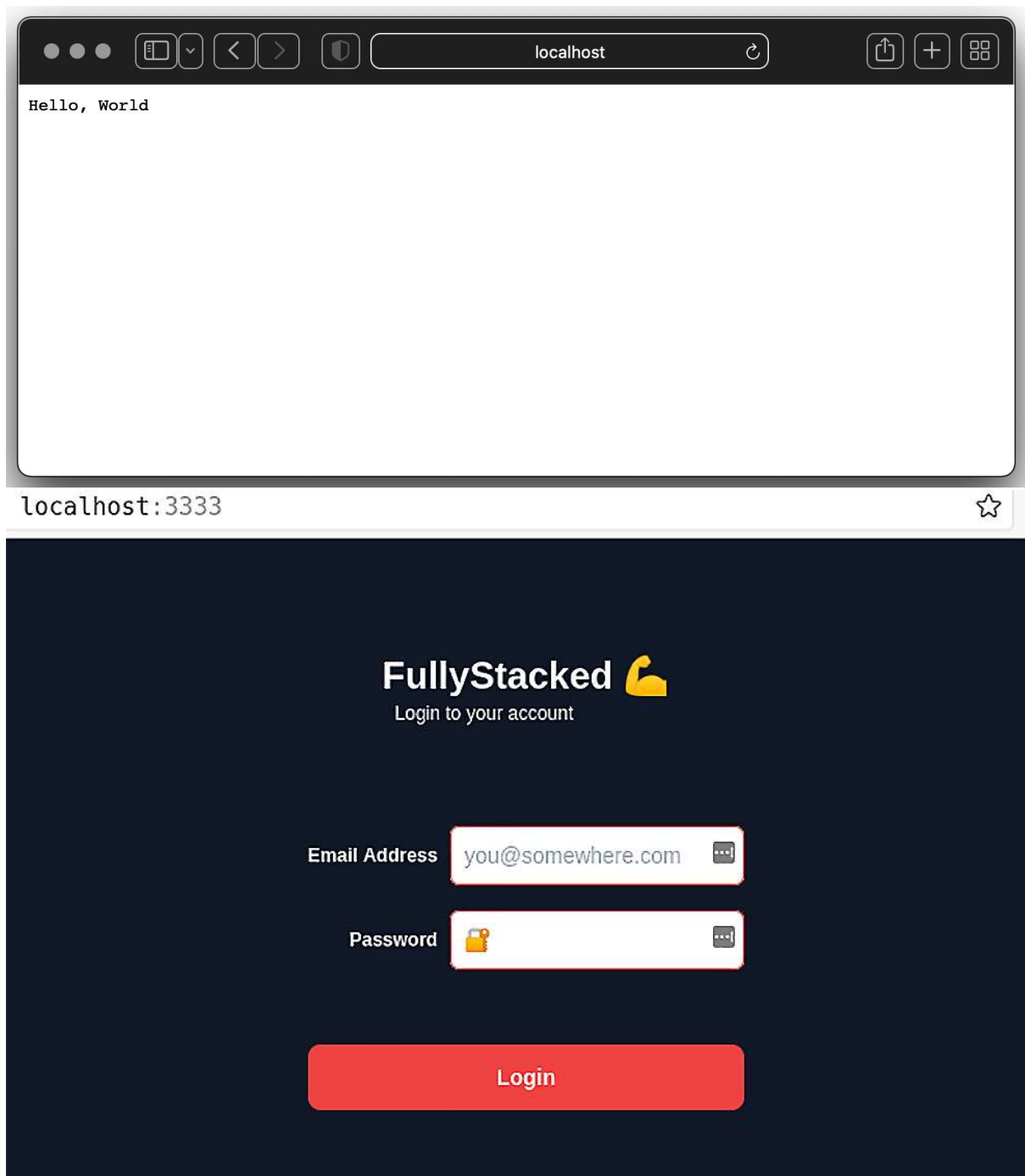
🔍 metric_random

metric_random gauge Random numbers

Table **Graph**



Chapter 4: Serving and Embedding HTML Content



localhost:3333/dashboard.html

FullyStacked 

Start Workout

Today's Workout

DAY 1, WEEK 1

Bench Press	2x5 1x5+	100kg
Bench Press	2x5 1x5+	100kg
Bench Press	2x5 1x5+	100kg

Workout on 2/1/2021

DAY 1, WEEK 1

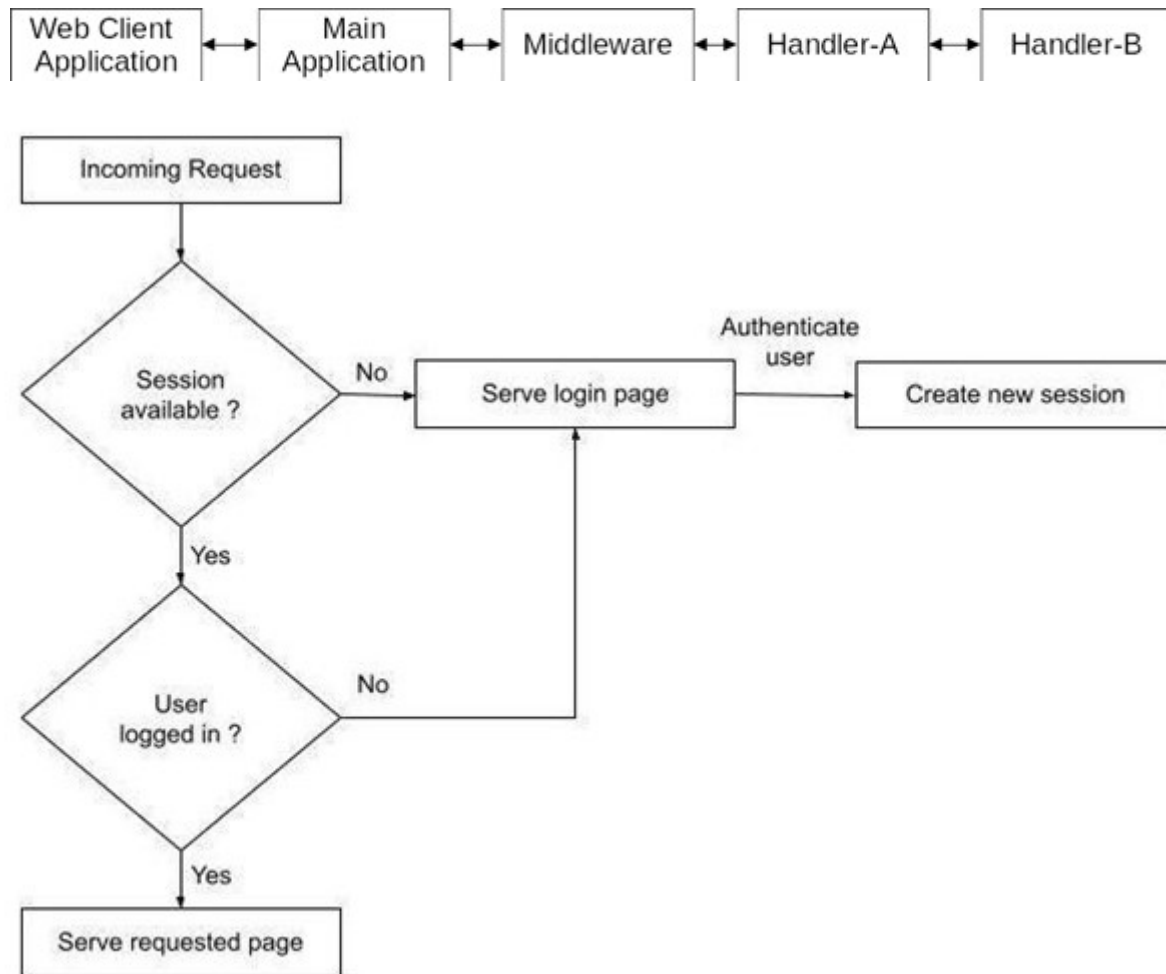
Bench Press	2x5 1x5+	100kg
Bench Press	2x5 1x5+	100kg
Bench Press	2x5 1x5+	100kg

No More Workouts

FullyStacked 

Login unsuccessful

Chapter 5: Securing the Backend and Middleware

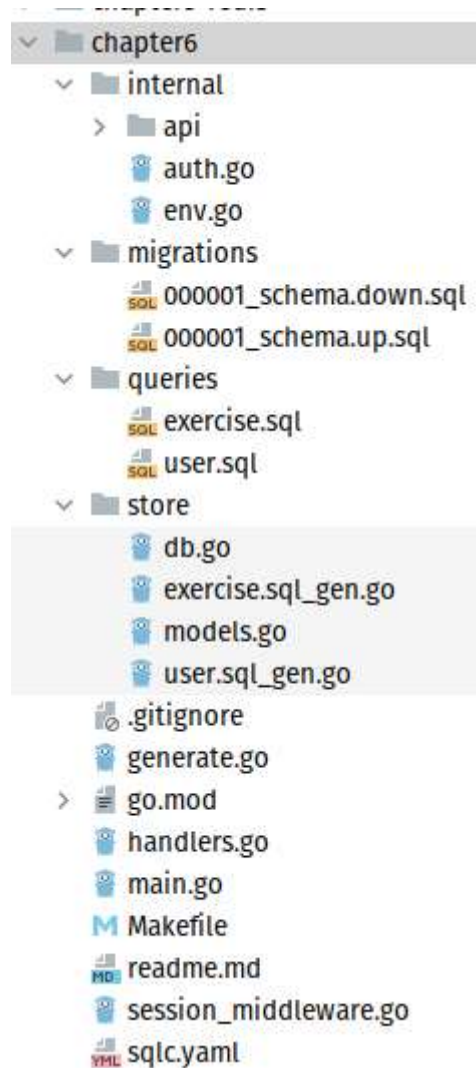


Request Headers (472 B)

Raw

```
Accept: image/webp,*/*
Accept-Encoding: gzip, deflate
Accept-Language: en-US,en;q=0.5
Connection: keep-alive
Cookie: session_token=MTY0MzIwMDQ00XxEdi1CQkFFQ180SUFBUkFCRUFBQUpmLUNBQUVHYzNSewFXNW5EQThBRFdG
MWRHaGxib1JwWTJGMFpXUUVZbT12YkFJQ0FBQT18f6UKeoMoR1M0fLWETjfQd3Yis1SGPiuk0Q_5brfFVuk=
DNT: 1
Host: localhost:3333
Referer: http://localhost:3333/login
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:89.0) Gecko/20100101 Firefox/89.0
```

Chapter 6: Moving to API-First



Client

Server

Preflight request

```
OPTIONS /doc HTTP/1.1
Origin: https://foo.example
Access-Control-Request-Method: POST
Access-Control-Request-Headers: X-PINGOTHER, Content-type
...
```

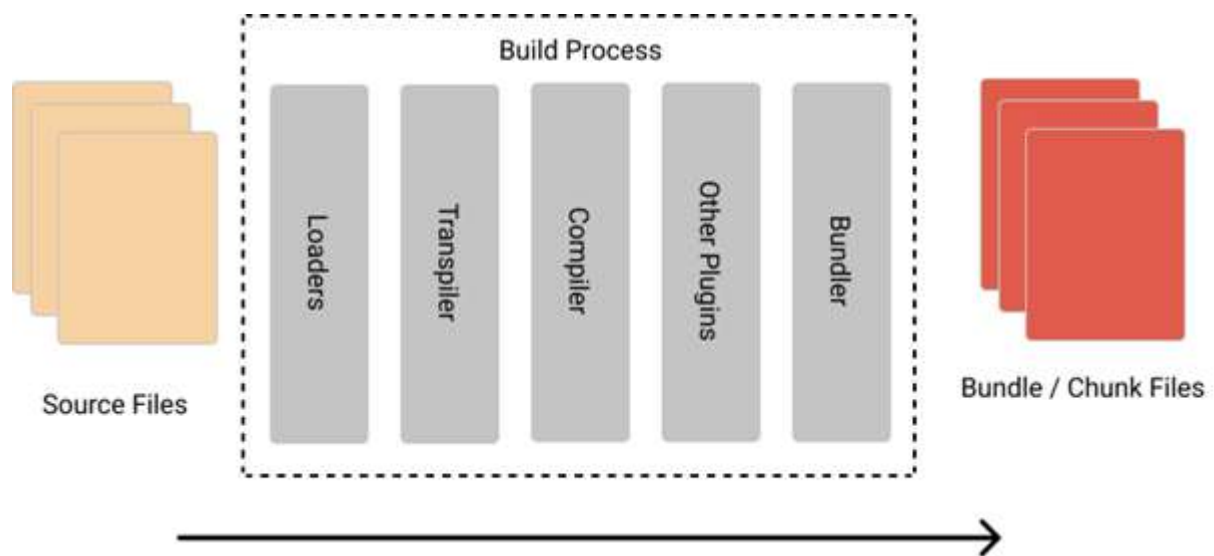
```
HTTP/1.1 204 No Content
Access-Control-Allow-Origin: https://foo.example
Access-Control-Allow-Methods: POST, GET, OPTIONS
Access-Control-Allow-Headers: X-PINGOTHER, Content-Type
Access-Control-Max-Age: 86400
...
```

Main request

```
POST /doc HTTP/1.1
X-PINGOTHER: pingpong
Content-Type: text/xml; charset=UTF-8
Origin: http://foo.example
...
```

```
HTTP/1.1 200 OK
Access-Control-Allow-Origin: http://foo.example
Vary: Accept-Encoding, Origin
Content-Encoding: gzip
Content-Length: 235
...
```

Chapter 7: Frontend Frameworks



The image shows a login form for 'FullyStacked'. The form has a dark blue background. At the top, it says 'FullyStacked' with a yellow heart icon and 'Login to your account'. Below this, there are two input fields: 'Email Address' with the placeholder text 'enter username' and 'Password' with the placeholder text 'enter password'. At the bottom, there is a red 'Login' button.

```
> test@1.0.0 start
> http-server ./ -p 3000

Starting up http-server, serving ./
Available on:
  http://127.0.0.1:3000
  http://192.168.1.3:3000
  http://172.17.0.1:3000
Hit CTRL-C to stop the server
□
```

```
vite v2.8.6 dev server running at:
```

```
> Local: http://localhost:3000/
```

```
> Network: use '--host' to expose
```

```
ready in 202ms.
```

Home Login

Home Screen

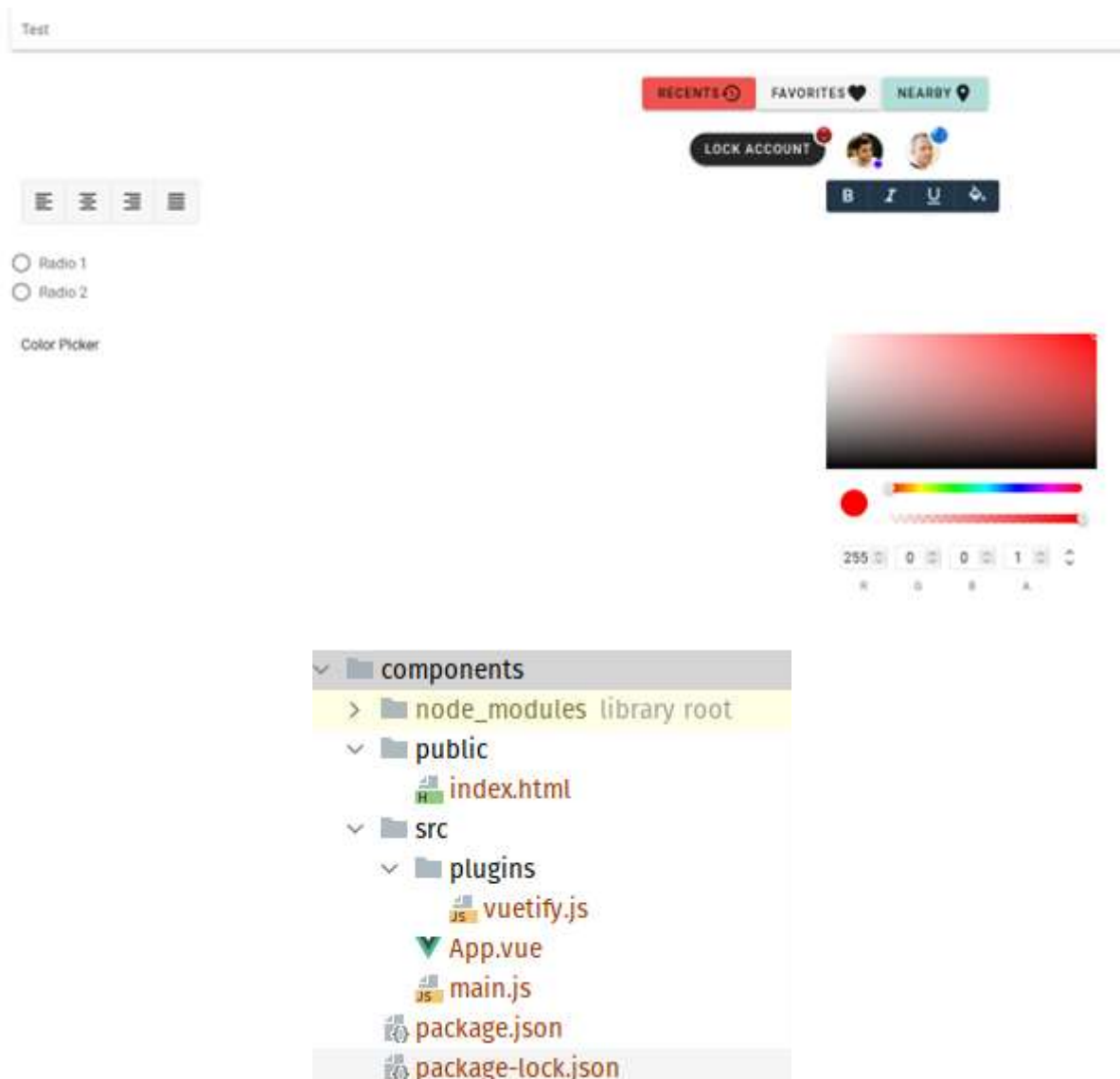
Chapter 8: Frontend Libraries

DONE Compiled successfully in 7784ms

App running at:

- Local: <http://localhost:8080/>
- Network: <http://192.168.1.5:8080/>

Note that the development build is not optimized.
To create a production build, run `npm run build`.



Hello Bulma!

Congratulations!

The page background should turn **green**!

Type	CSS class
Columns	<code>columns</code> <code>column</code>
Layout	<code>section</code> <code>container</code> <code>footer</code>
Elements	<code>button</code> <code>content</code> <code>title</code> <code>subtitle</code>
Form	<code>field</code> <code>control</code>
Helpers	<code>has-text-centered</code> <code>has-text-weight-semibold</code>

[Bulma homepage](#)[Documentation](#)

Slide 5

ID	First Name	Last Name	Date	Gender
1	Jerick	Simmone	2016-10-15 13:43:27	Male
2	John	Jacobs	2016-12-15 06:00:53	Male
3	Tina	Gilbert	2016-04-24 04:26:38	Female
4	Clarence	Flores	2016-04-10 10:28:40	Male
5	Anne	Lee	2016-12-06 14:38:38	Female

Right Top (default) Bottom Left Default

Home / Docs / Breadcrumb

Click me!

Subtitle

Lorem ipsum dolor sit amet, consectetur adipiscing elit.
 Nulla ac ipsum, nulla ultricies eleifend gravida, nulla nunc varius lectus, nec rutrum justo nibh eu lectus.
 Ut volutpat semper dui. Prae erat odio, efficitur vel erat vel, interdum nulla neque.

The Email Address must be at least 6 characters.

The Email Address must be a valid email.

Email Address

email: nick

nick@bar.com

```
{
  "text": {
    "required": true,
    "email": true,
    "minLength": true,
    "$model": "nick@bar.com",
    "$invalid": false,
    "$dirty": true,
    "$anyDirty": true,
    "$error": false,
    "$anyError": false,
    "$pending": false,
    "$params": {
      "required": {
        "type": "required"
      },
      "email": {
        "type": "email"
      },
      "minLength": {
        "type": "minLength",
        "min": 2
      }
    }
  },
  "$model": null,
  "$invalid": false,
  "$dirty": true,
  "$anyDirty": true,
  "$error": false,
  "$anyError": false,
  "$pending": false,
  "$params": {
    "text": null
  }
}
```

3412 212112 11121



American Express: starts with 34/37

34●● ●●●●●● ●●●●●●

4242 4222 4242 4244

4242 4222 4242 4244 visa

Chapter 9: Tailwind, Middleware, and CORS

Button



Hello Vue 3 + Vite

Recommended IDE setup: [VSCode](#) + [Volar](#)

[Vite Documentation](#) | [Vue 3 Documentation](#)

count is: 0

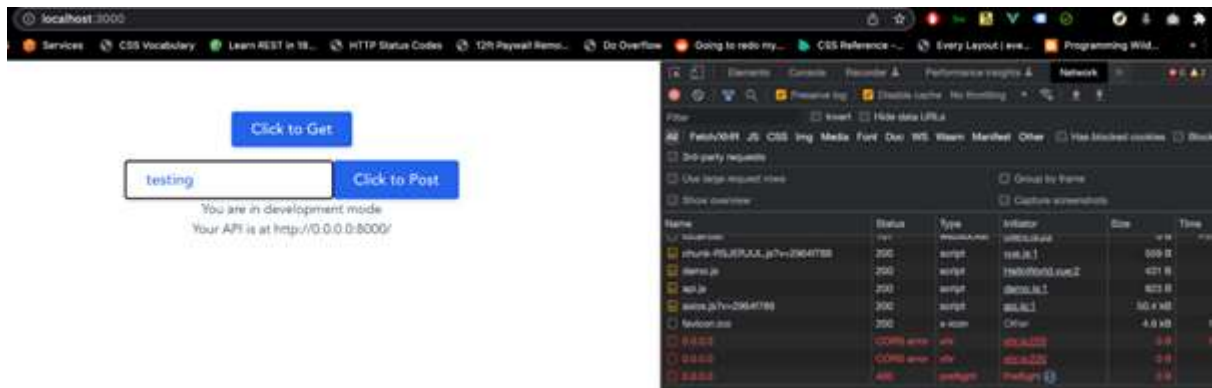
Edit components/HelloWorld.vue to test hot module replacement.

Click me - my count is 3

Click to Get

Click to Post

You are in development mode
Your API is at http://0.0.0.0:8000/



Click to Get

eMesreveR

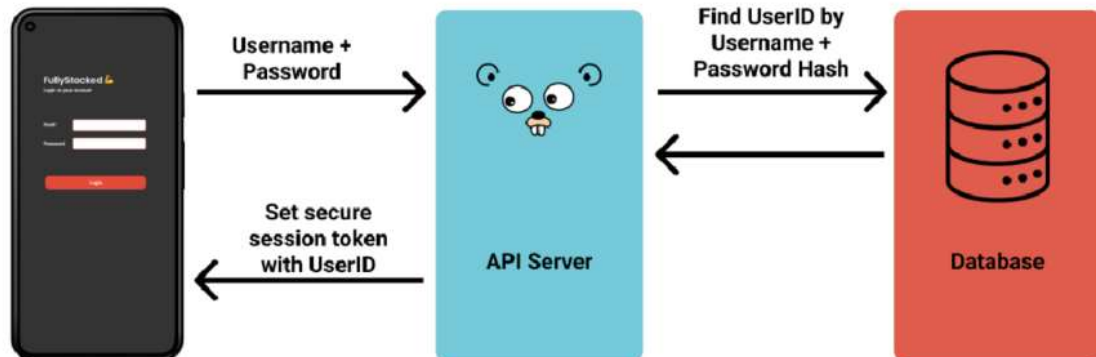
Click to Post

You are in development mode
Your API is at http://0.0.0.0:8000/

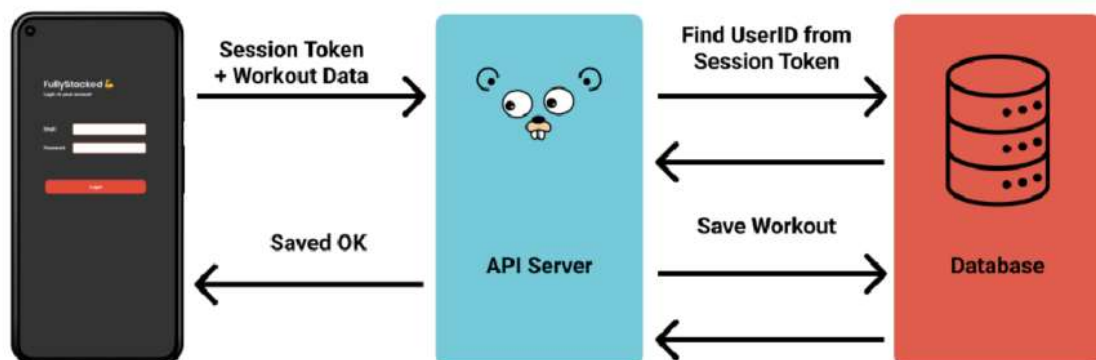
- gnitset
- Hello World
- eMesreveR
- ReverseMe

Chapter 10: Session Management

Login



Save Workout



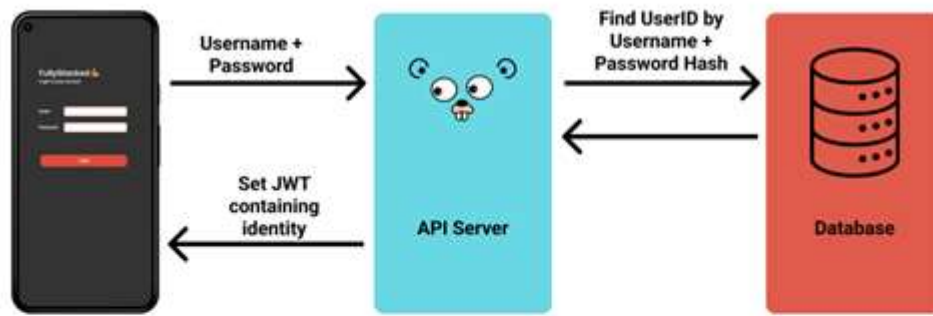
```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMjM0NTY3ODkwIiwidGUiOiJGdWxsfQ.Bph4Comr-law35-d1kJSDS3riNiS7GsF2NwJNB8I-0M
```

Header - algorithm used and the token type

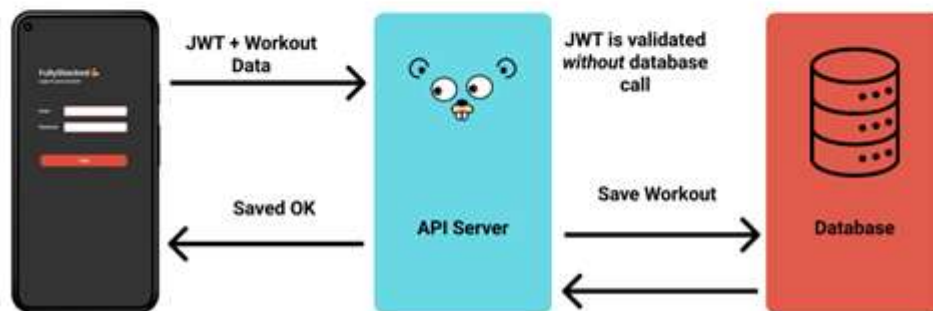
Payload - The data and claims

Signature - Used to verify authenticity

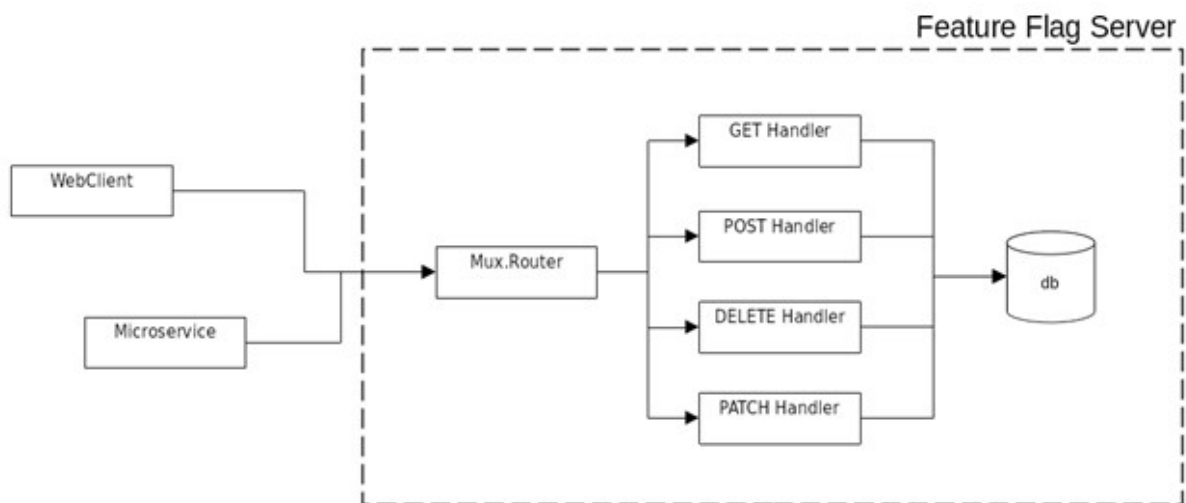
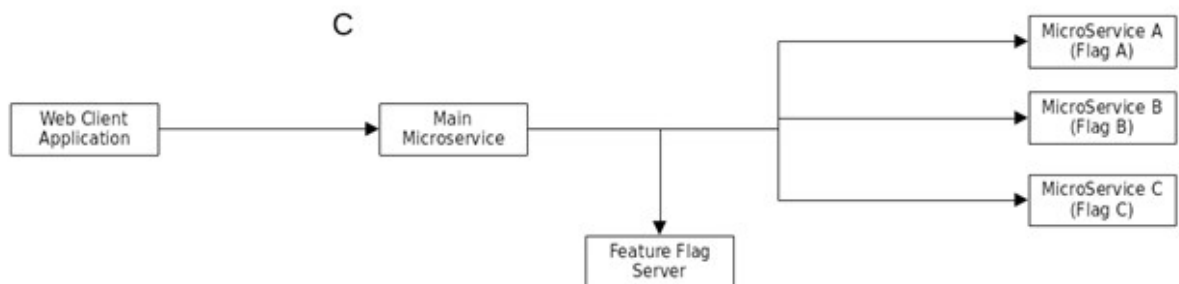
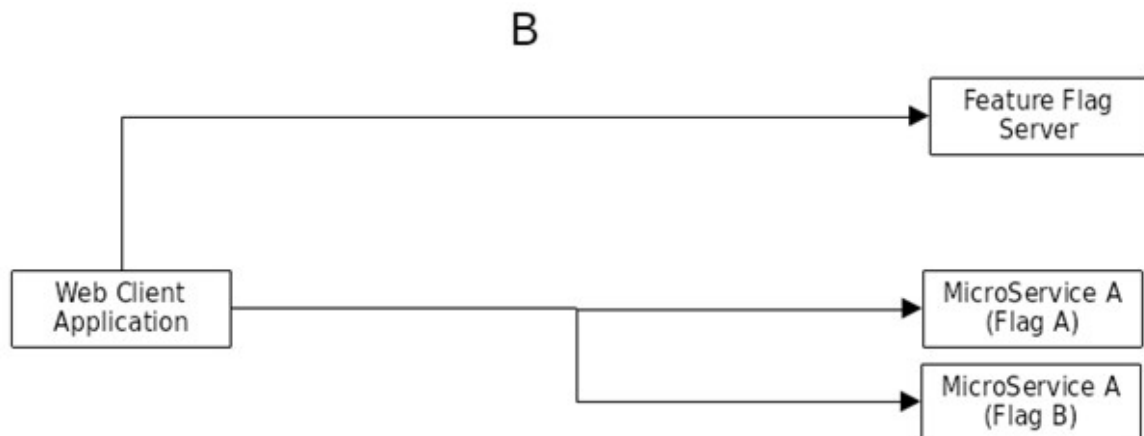
Login

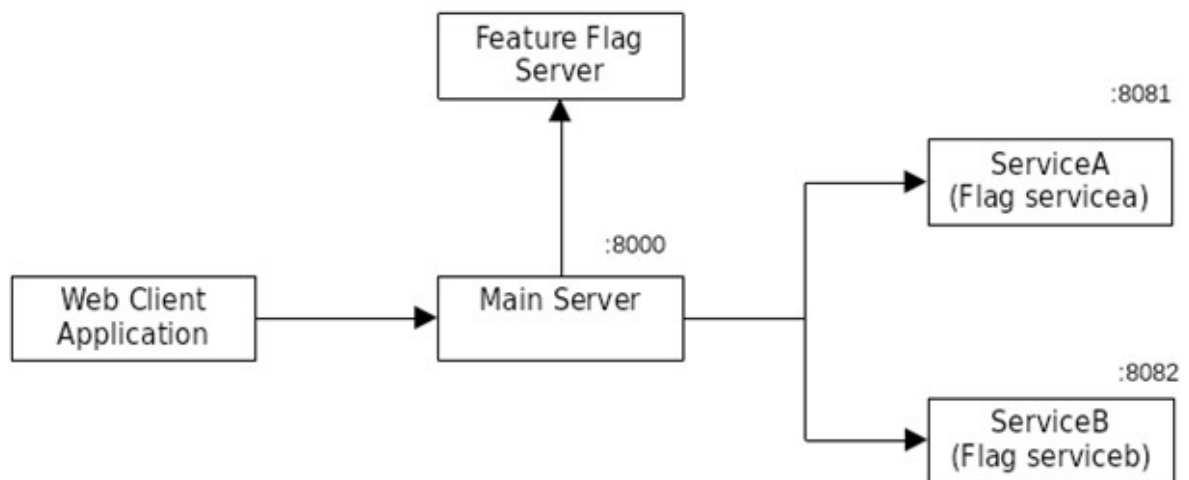
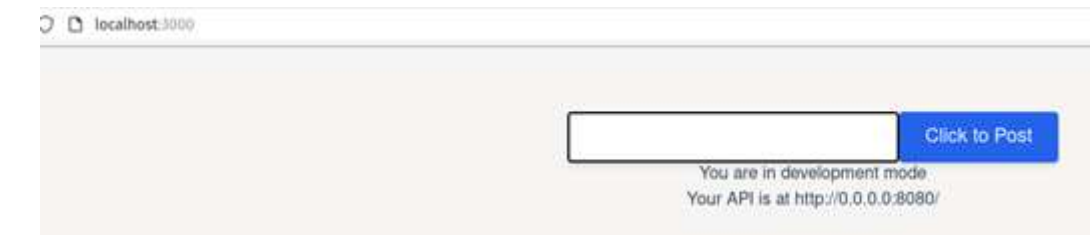


Save Workout



Chapter 11: Feature Flags





Chapter 12: Building Continuous Integration

nanikjava/golangci

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Quick setup — if you've done this kind of thing before

or HTTPS SSH git@github.com:nanikjava/golangci.git

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [github](#).

...or create a new repository on the command line

```
echo "# golangci" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin git@github.com:nanikjava/golangci.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin git@github.com:nanikjava/golangci.git
git branch -M main
git push -u origin main
```

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Get started with GitHub Actions

Build, test, and deploy your code. Make code reviews, branch management, and issue triaging work the way you want. Select a workflow to get started.

[Skip this and set up a workflow yourself](#) →

Search workflows

golangci / golangci

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

golangci / golangci / workflows / main.yml

Cancel changes Start commit

Build new file

Preview

Issues 0 0 0 No merge 0

```
1 # This is a basic workflow to help you get started with Actions
2
3 name: CI
4
5 # Controls when the workflow will run
6 on:
7   # Triggers the workflow on push or pull request events but only for the "main" branch
8   push:
9     branches: [ "main" ]
10  pull_request:
11    branches: [ "main" ]
12
13 # Allows you to run this workflow manually from the Actions tab
14 workflow_dispatch:
15
16 # A workflow run is made up of one or more jobs that run sequentially or in parallel
17 jobs:
18   # This workflow contains a single job called "build"
19   build:
20     # The type of runner that the job will run on
21     runs-on: ubuntu-latest
22
23     # Steps represent a sequence of tasks that will be executed in order of the job
```

Marketplace Documentation

Search Marketplace for Actions

Featured Actions

Cache

By actions

Cache artifacts, dependencies, and build outputs to improve workflow execution time

Setup Node.js environment

By actions

Setup a Node.js environment by setting runtime, matrix, and optionally downloading and adding it to the PATH

Setup Go environment

By actions

Setup a Go environment and add it to the PATH

Close State Issues

By actions

[Code](#) [Issues](#) [Pull requests](#) [Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

Create main.yaml GitHub Actions Demo #1

Summary

Jobs

Explore-GitHub-Actions

Explore-GitHub-Actions

Automated 24 minutes ago 1/16

1

Set up job

2

Run echo: "👋 The job was automatically triggered by a push event."

3

Run echo: "📍 This job is now running on a Linux server hosted by GitHub"

4

Run echo: "📁 The name of your branch is refs/heads/main and your repository is henkjanvelling1."

5

Check out repository code

6

Run echo: "📦 The henkjanvelling1 repository has been cloned to the runner."

7

Run echo: "🚀 The workflow is now ready to run your code in the future!"

8

List files in the repository

9

Run echo: "🟢 This job's status is success."

10

Post-Check out repository code

11

Complete job

Add file  Code

20 hours ago 1 commit

20 hours ago

20 hours ago

20 hours ago

20 hours ago

20 hours ago

20 hours ago

20 hours ago

20 hours ago

20 hours ago

20 hours ago

20 hours ago

About

No description, website, or topics provided.

0 stars

1 watching

0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

 Clear current search query, filters, and sorts

0 packages



No results matched your search.

Try [browsing all packages](#) to find what you're looking for.

⚙ General

Access

👤 Collaborators

🔒 Moderation options ▾

Code and automation

🔗 Branches

🏷 Tags

▶ Actions ▲

General

Runners

⚙ ...

Actions permissions

- ☒ **Allow all actions and reusable workflows**
Any action or reusable workflow can be used, regardless of the repository's permissions.
- ☐ **Disable actions**
The Actions tab is hidden and no workflows can run.
- ☐ **Allow nanikjava actions and reusable workflows**
Any action or reusable workflow defined in a repository.
- ☐ **Allow nanikjava, and select non-nanikjava, actions and reusable workflows**
Any action or reusable workflow that matches the specified criteria. [more about allowing specific actions and reusable workflows](#)

Save

Workflow permissions

Choose the default permissions granted to the GITHUB_TOKEN when running workflows in this repository. You can specify more granular permissions in the workflow using YAML. [Learn more](#).

- ☒ **Read and write permissions**
Workflows have read and write permissions in the repository for all scopes.
- ☐ **Read repository contents permission**
Workflows have read permissions in the repository for the contents scope only.

Choose whether GitHub Actions can create pull requests or submit approving pull request reviews.

- ☒ **Allow GitHub Actions to create and approve pull requests**

Save

Initial commit Build Docker Image #1

Summary

Jobs

Push Docker image to Github Packages

Push Docker image to Github Packages

successful now in 80s

- > ☒ Set up job
- > ☒ Checkout repository
- > ☒ Log in to the Container registry
- > ☒ Build and push Docker image
- > ☒ Post Build and push Docker image
- > ☒ Post Log in to the Container registry
- > ☒ Post Checkout repository
- > ☒ Complete job

chapter12

Install from the command line

Learn more

```
$ docker pull ghcr.io:jamendajee/jee/chapter12:latest
```

Recent tagged image versions

latest

1

Published 2 minutes ago · Digest: sha256:...

[View and manage all versions](#)

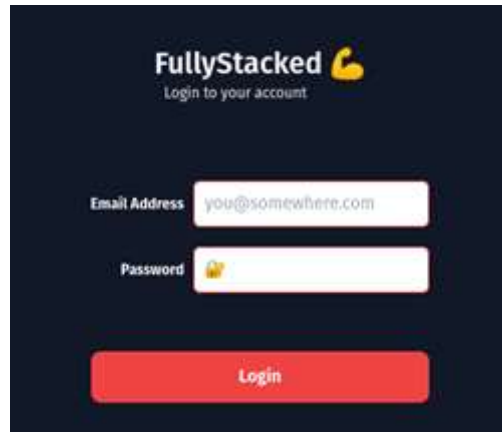


Add a Readme to the linked repository

The linked repository's Readme will be shown here.

Chapter 13: Dockerizing an Application

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
redis	latest	bba24acba395	4 weeks ago	113MB
postgres	latest	1ee973e26c65	4 weeks ago	376MB



Chapter 14: Cloud Deployment

Global Network of AWS Regions

The AWS Cloud spans 87 Availability Zones within 27 geographic regions around the world, with announced plans for 21 more Availability Zones and 7 more AWS Regions in Australia, Canada, India, Israel, New Zealand, Spain, and Switzerland.



Products / Compute / Amazon EC2 / ...

Instance Type Explorer

Amazon EC2 provides a wide selection of over 400 instance types optimized to fit different use cases. With EC2 Instance Type Explorer, you can more easily navigate and discover the instances that are right for your workloads and business needs. Using filters, you can quickly narrow the search for the right instance family based on instance category, or by hardware configuration. For additional details on each instance family, visit [EC2 Instance Types Details Page](#) or respective product details page.

[Clear filters](#)

Instance Categories

▼ All Instance Categories

- ☐ General Purpose
- ☐ Compute Optimized
- ☐ Memory Optimized
- ☐ Storage Optimized
- ☐ Accelerated Computing

Hardware Configuration

▼ Processors

- ☐ AWS Graviton
- ☐ AMD
- ☐ Intel

▼ Accelerators

- ☐ AWS Inferentia
- ☐ NVIDIA GPU
- ☐ AMD GPU

Search instances

1-12 (54)

COMPUTE OPTIMIZED

Amazon EC2 C6a Instances

Amazon EC2 C6a instances are powered by 3rd generation AMD EPYC processors and are designed for compute-intensive workloads.

Ideal use cases:

Compute-intensive workloads such as batch processing, distributed analytics.

GENERAL PURPOSE

Amazon EC2 M6i Instances

Amazon EC2 M6i instances are powered by 3rd generation Intel Xeon Scalable processors that offer a balance of compute, memory, and networking resources for a broad range of workloads.

Ideal use cases:

GENERAL PURPOSE

Amazon EC2 Mac Instances

Mac instances are powered by Apple Mac mini computers and built on the AWS Nitro System. This EC2 family gives developers access to macOS so they can develop, build, test, and sign applications that require the Xcode IDE.

Ideal use cases:

Developing, building, testing, and signing iOS, iPadOS, macOS, watchOS, and tvOS applications on the Xcode IDE.

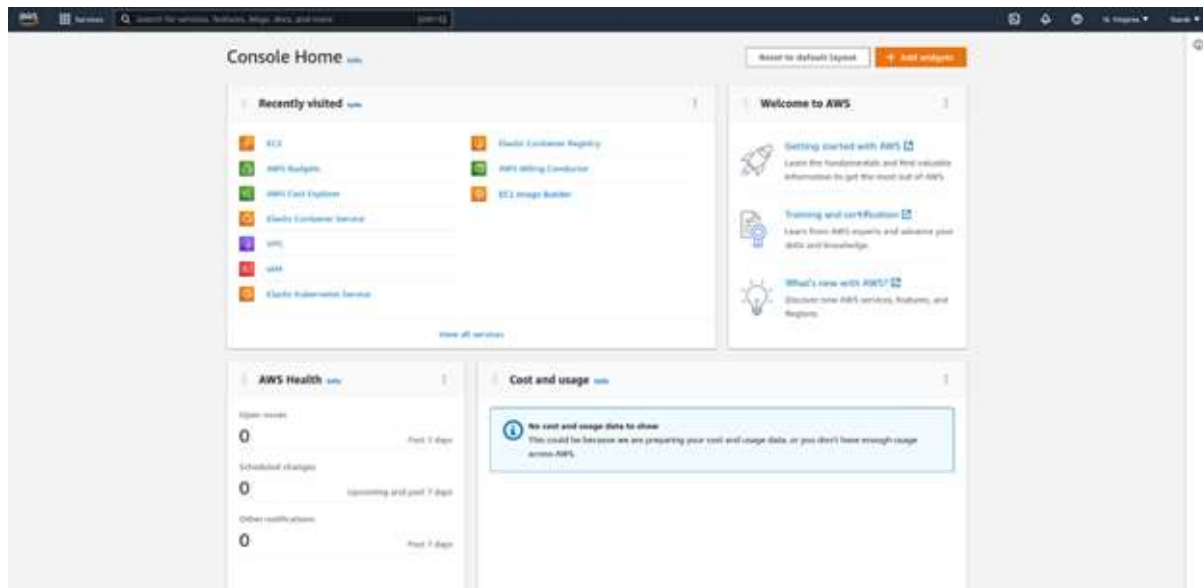
Key technology:

- Intel core i7 processors with 3.2 GHz (4.8 GHz turbo)
- 6 physical / 12 logical cores

GENERAL PURPOSE

COMPUTE OPTIMIZED



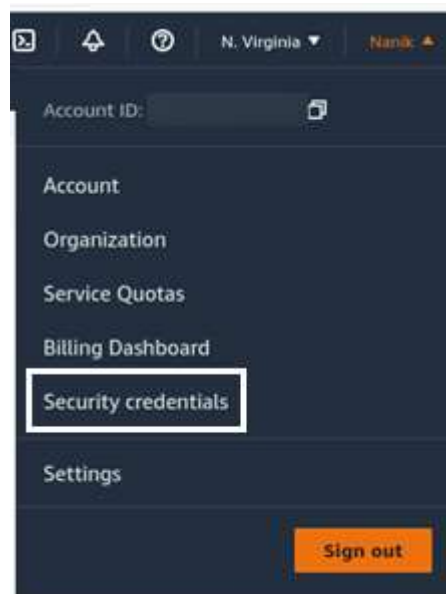


Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.



Your Security Credentials

Use this page to manage the credentials for your AWS account. To manage credentials for AWS Identity and Access Management (IAM) users, use the [IAM Console](#).

To learn more about the types of AWS credentials and how they're used, see [AWS Security Credentials](#) in *AWS General Reference*.

- Password
- Multi-factor authentication (MFA)
- Access keys (access key ID and secret access key)

Use access keys to make programmatic calls to AWS from the AWS CLI, tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time.

For your protection, you should never share your secret keys with anyone. As a best practice, we recommend frequent key rotation.

If you lose or forget your secret key, you cannot retrieve it. Instead, create a new access key and make the old key inactive. [Learn more](#).

Created	Access Key ID	Last Used	Last Used Region	Last Used Service	Status	Actions
			us-east-1	S3	Active	Make inactive Delete

AWS

